

Supplementary Methods:

LD-based complexity reduction and the analysis of ancestry sorting

A recurring principle organises the procedures below. Markers in linkage disequilibrium (LD) do not carry independent information, so the unit of analysis throughout is the LD cluster rather than the marker; quantities are measured per cluster, and statistical tests treat clusters, not markers, as observations. Two further conventions follow from the same reasoning and are stated where they arise: a variable that is to be tested as a predictor is never also used to select the data, and a criterion whose stringency would otherwise depend on cluster size is expressed as a rate rather than a count.

S1 Data

S1.1 Samples

Twenty hybrid populations between *F. polyctena* and *F. aquilonia* were sampled (164 individuals in total), together with allopatric reference individuals of both parental species. *Formica* wood ants are haplodiploid, so males are haploid; this is relevant to the demographic modelling reported separately and to the interpretation of within-population variation.

S1.2 Markers and ancestry polarisation

Genotypes and per-marker ancestry information were taken from the DIEM output for this dataset. Sites were restricted to those called as biallelic. For each marker DIEM supplies a polarity and a diagnostic index (DI), the latter measuring how informative the marker is about parental ancestry. Genotype dosages at markers whose polarity indicated the opposite orientation were replaced by 2 – dosage, so that after this step dosage is on a single, consistent ancestry-oriented scale genome-wide. All later use of DI refers to this per-marker diagnostic index.

S1.3 Frequency filtering and the two genotype matrices

Minor allele frequencies were computed among the hybrid individuals only, and markers retained at $MAF \geq 0.05$, giving 1,114,340 markers. Two genotype matrices were then constructed over this same marker set. The first contains the 164 hybrid individuals only and is used for all LD estimation, clustering and complexity reduction. The second additionally contains the parental reference individuals and is used only where parental allele frequencies are required, for example to orient ancestry or to compute between-species divergence. The separation is deliberate: parental and hybrid individuals differ far more from each other than hybrids differ among themselves, so including parents in the LD computations would let between-species differentiation dominate every pairwise LD estimate and obscure the within-hybrid-zone LD structure that the clustering is designed to capture.

S2 Estimation of LD decay

S2.1 Model and fitting

LD decay was modelled per chromosome as

$$r^2(d) = b + \frac{c - b}{1 + a d}, \quad (1)$$

where d is physical distance in base pairs, a is the decay rate, b is the long-range background level of r^2 , and c is short-range r^2 . Background LD was estimated from inter-chromosomal marker pairs, which are unlinked by construction and therefore provide the appropriate baseline. Within each chromosome the curve was fitted across twenty sliding windows with 50% overlap, sampling up to 5,000 marker pairs per window stratified across twenty distance classes so that short and long distances are represented comparably; r^2 was summarised at its 0.95 quantile, and the per-window fits were aggregated over the central 95% of windows to limit the influence of individual outlying windows.

S2.2 Allele-frequency filtering during fitting

Only pairs in which both markers exceeded MAF 0.1 contributed to the background and decay-curve fits. Pairs involving rare alleles mechanically depress r^2 irrespective of true linkage, and including them biases the fitted a , b and c . This restriction applies to curve fitting alone: the stored pairwise LD edge lists retain all markers, so no marker is lost from the downstream clustering.

S2.3 Decay rate, chromosome size, and window sizes

Decay rates vary systematically with chromosome length, so a robust regression of a on chromosome size was fitted genome-wide, allowing decay behaviour to be described consistently across chromosomes of very different length. For a target relative LD threshold ρ , the corresponding physical distance follows from the fitted curve as

$$d = \frac{\rho}{a(1 - \rho)}, \quad (2)$$

which is how all distance-based windows below are obtained. Anchoring distances in each chromosome's own fitted decay curve, rather than in a fixed base-pair window or a fixed r^2 cutoff, is what makes a single threshold behave comparably across chromosomes with different background LD.

S2.4 Local LD support

For each marker a measure of local LD support, ld_w , was computed as the median r^2 between that marker and its neighbours within the window corresponding to $\rho = 0.95$. It summarises how strongly a marker is embedded in local LD, and is used below to identify the regions where Stage 1 clustering is most likely to have been conservative.

S3 Complexity reduction: one partition, two projections

S3.1 Rationale

LD pruning and the construction of consensus genotypes are usually treated as separate procedures, but they answer the same question: what is the finest partition of markers that can no longer be distinguished statistically, and therefore constitutes an independent unit in these data? Once that partition is fixed, two projections of it are useful. Selecting one representative marker per cluster gives an LD-pruned marker set. Collapsing each cluster to a single consensus dosage gives an expected multi-locus genotype (eMLG) per cluster. Both are produced here from a single clustering pass, so the two representations always refer to the same underlying units.

S3.2 What a cluster represents, and why it is not a haplotype block

A biallelic marker carries exactly one bipartition of the sampled chromosomes: it divides them into two groups and can encode nothing further. Where the local genealogy contains more than two segregating lineages, mutations falling on different branches define *different* bipartitions. Markers arising on the same branch are carried by the same set of lineages and are therefore perfectly correlated, in the absence of recombination, recurrent mutation and missing data; markers arising on different branches are correlated only partially, according to how those branches are related.

Clustering on linkage disequilibrium therefore groups markers by the partition of individuals they report, not by their position, and two consequences follow. First, clusters need not partition the chromosome: where several branches segregate in one region, the markers tagging each are physically interleaved, so the resulting clusters overlap along the sequence. The distance restriction applied below keeps clusters local, but locality is not spatial exclusivity. Second, because r^2 is sensitive to allele frequency, two markers tagging *nested* branches can show complete association on a frequency-independent measure and yet a low r^2 ; clustering on r^2 accordingly separates nested branches, which is precisely the situation in which a single interval contains several overlapping clusters.

For these reasons we avoid the term haplotype block. A block, in its usual sense, is a contiguous stretch of limited haplotype diversity with defined boundaries, implicitly reflecting a single genealogy, and it partitions the sequence. The units here are instead sets of markers reporting the same partition of individuals, several of which may occupy the same interval. The reduction performed below is over information content—which split of the sample a marker reports—rather than over physical space.

This is also what makes a consensus genotype meaningful. Averaging markers that report the same partition reinforces the dosage of that partition, whereas averaging across partitions would blur distinct splits. The complete-linkage requirement in Stage 1 and the pairwise correlation floor in Stage 2 both exist to prevent that.

S3.3 Stage 1: local LD clusters

The clustering threshold was derived, per chromosome, from that chromosome’s own fitted decay curve at $\rho = 0.5$, its half-decay point, rather than from a fixed r^2 cutoff applied genome-wide. The threshold was deliberately not allowed to vary further *within* a chromosome: a region with unusually slow local decay, such as a centromere or an inversion, is precisely where markers are most redundant and pruning should be most aggressive, so adapting the threshold to tolerate high local LD there would defeat the purpose.

Clustering proceeded in two steps on the precomputed pairwise LD edge list. Markers were first grouped into connected components of the thresholded sparse graph, which is inexpensive; each component was then refined into complete-linkage sub-clusters by hierarchical clustering on the dense sub-matrix of r^2 values within it. The second step is necessary because single linkage alone permits chaining through a mosaic LD pattern: two markers that are uncorrelated with each other can be joined through an intermediate marker correlated with both, and would then be wrongly treated as redundant. Complete linkage instead requires every pairwise r^2 within a final cluster to clear the threshold.

The representative of each cluster was taken to be the marker with the highest median r^2 to the remaining cluster members. This is preferred to selecting the marker most strongly correlated with the cluster’s mean genotype, because the latter is sensitive to allele-coding direction: a genuinely central marker coded on the opposite allele correlates negatively with the mean and would be passed

over. Since r^2 ignores sign, the median- r^2 criterion does not have this failure mode.

S3.4 Why Stage 1 is conservative, and how Stage 2 resolves it

The pairwise LD edge list is built within a sliding window, so pairs of markers farther apart than that window were never evaluated; in the clustering they behave as though $r^2 = 0$. Single linkage tolerates this, because a chain of overlapping windows still connects distant markers. Complete linkage does not: a genuine single cluster wider than the window is fragmented into several clusters purely because the unevaluated pairs default to zero. The window is sized from average decay behaviour to reach a high ρ , which covers typical regions comfortably, but slowly-recombining regions decay far more slowly than that average and are where fragmentation is expected.

Stage 2 addresses this after the fact rather than by changing Stage 1. Candidate clusters are compared directly from genotypes, with no window restriction, and are selected for reconsideration using Stage 1's own cluster boundaries. Markers are deliberately not pre-split by their local LD before clustering: doing so severs real clusters exactly at the selection boundary, whereas selecting whole clusters leaves genuine clusters intact.

S3.5 Stage 2: quality-gated merging

A Stage-1 cluster was flagged for reconsideration if any of its members exceeded $\text{ld}_w > 0.025$, or if the cluster contained at least five markers. The second criterion captures substantial clusters whose local LD support is moderate and which would otherwise pass through unmerged. Flagged clusters were then merged by a distance-restricted, average-linkage dynamic tree cut that accepts each candidate merge only while two quality conditions hold: the consensus genotype must survive hard-calling, quantified as $\text{score}_{\text{eMLG}}(x) = \text{cor}(\text{round}(x), x)^2 \geq 0.80$, which is the property downstream LD statistics require of a consensus; and the two sides being merged must themselves be correlated, at pairwise $r^2 \geq 0.2$. The two conditions are not redundant. Fidelity is a property of the resulting consensus, and once a cluster is large that consensus is dominated by the markers already in it, so appending a few further markers changes the score very little whether or not they report the same partition. The fidelity criterion is therefore least discriminating precisely where clusters are largest, and the pairwise requirement is what prevents an essentially uncorrelated marker from being absorbed there. Because merging proceeds only while both conditions hold, this stage cannot over-merge regardless of how permissively clusters were flagged.

The distance restriction was applied in genetic rather than physical units, at 0.5 cM, using the linkage map. Physical distance is a poor and sometimes misleading proxy for recombination distance in this genome: cluster pairs more than a megabase apart can be at essentially zero genetic distance, while others only a few hundred kilobases apart span several centimorgans. A genetic threshold also behaves correctly on exceptional chromosomes, where a slower local recombination rate gives 0.5 cM a proportionally larger physical span, instead of clipping a genuine cluster to a distance calibrated on typical chromosomes.

S3.6 Outputs and conventions

The procedure returns both projections of the partition: one representative marker per cluster, and a consensus dosage for clusters of at least five markers ($\text{min_n_loci_eMLG} = 5$). The size threshold is a computational limit on consensus construction, not a redefinition of the unit: clusters below it remain valid units and retain their pruning representative, and the pruned marker set is unchanged by this setting. Clustering results were written to a file whose name encodes the parameters used, so that analyses depending on a particular clustering cannot silently be run against a different one.

S4 Sorting statistics

S4.1 Orientation and near-fixation

The question these statistics address is whether parental ancestry sorts *predictably* across the replicate hybrid populations—the same parental allele approaching fixation repeatedly—or in directions that vary among populations. For each locus, dosage was oriented into an *F. aquilonia*-allele frequency using the parental reference samples: the sign of the difference in allele frequency between the two parental species defines which allele is counted, and loci at which the parents do not differ have no defined orientation. Each hybrid population was then classified as near-fixed toward *F. aquilonia* if its oriented frequency was at least $1 - \theta$, near-fixed toward *F. polycтена* if it was at most θ , and unsorted otherwise, with $\theta = 0.15$. This tolerance interacts with population sample size, since a population of few diploid individuals can only realise a coarse grid of frequencies, and was therefore held constant across all analyses rather than tuned per analysis.

S4.2 Decomposition and classification

Let n_{obs} be the number of populations with data and a defined orientation, and n_{aqu} and n_{pol} the numbers near-fixed in each direction. Sorting was decomposed as

$$\underbrace{\frac{n_{\text{aqu}} + n_{\text{pol}}}{n_{\text{obs}}}}_{\text{prop_fixed}} = \underbrace{\left| \frac{n_{\text{aqu}} - n_{\text{pol}}}{n_{\text{obs}}} \right|}_{\text{uni}} + \underbrace{\frac{2 \min(n_{\text{aqu}}, n_{\text{pol}})}{n_{\text{obs}}}}_{\text{bi}}, \quad (3)$$

separating the extent to which populations fixed in a consistent direction from the extent to which they fixed in opposing directions. A single threshold $\tau = 0.5$ on these quantities classifies each locus: unidirectional (toward one parent) when $|\text{uni}| \geq \text{bi}$ and $|\text{uni}| \geq \tau$, bidirectional when $\text{bi} > |\text{uni}|$ and $\text{bi} \geq \tau$, and unsorted otherwise. Because all three quantities are fractions of the population panel, the same threshold is directly comparable between individual markers and aggregated clusters, and genome-wide summaries are simple tallies over units.

S4.3 Test of directional predictability

Among the n_{fixed} populations that near-fixed in either direction, predictability was tested as

$$n_{\text{aqu}} \sim \text{Binomial}(n_{\text{fixed}}, 0.5), \quad (4)$$

two-sided and exact, computed only where $n_{\text{fixed}} \geq 5$, with Benjamini–Hochberg adjustment across tested loci. The null probability of 0.5 encodes the symmetric expectation, under which populations fix in either direction with equal chance. A per-locus alternative, in which each locus is tested against its own mean ancestry across the hybrid populations, is not appropriate here: that expectation is itself a direct function of the outcome being tested, so it absorbs the very signal of interest. Genome-wide mean *F. aquilonia* ancestry at diagnostic loci is close to 0.5, so the symmetric null is also the empirically correct one.

The descriptive classification and the test are reported as separate quantities throughout. The classification states how much and in which direction a unit sorted, at a fixed threshold; the test states whether the direction departs from the random-direction expectation. They answer different questions and are never combined into a single label.

S4.4 The polymorphism gate

Near-fixation in the hybrid populations is evidence of *sorting* only if the locus was segregating in the founding admixture. Analyses were therefore restricted to loci polymorphic in the parents, using the pooled parental minor allele frequency—computed from pooled allele counts across all parental individuals and folded to the minor allele—with a threshold of 0.15.

The gate is deliberately placed on parental allele frequency rather than on the diagnostic index. DI is a predictor to be tested in later analyses, against recombination, cluster size, climate association and the direction of sorting. Selecting loci on DI would truncate its range, attenuating every one of those relationships, and would make the question of whether DI predicts sorting circular. Parental allele frequency instead performs the role a gate should perform here: it removes loci that were already near-monomorphic in the founding pool, where near-fixation in the hybrids reflects the starting condition rather than a sorting event, while leaving DI free to vary across its full range.

We note that the magnitude of sorting and the unsorted/bidirectional/ unidirectional classification are invariant to which allele is designated *F. aquilonia*, since exchanging the labels exchanges n_{aqu} and n_{pol} without altering their sum or their minimum. Only the parental *direction* depends on the orientation, and is therefore interpretable only where the parental samples differ.

S4.5 Testing at the level of clusters

The same statistic was applied to the eMLG consensus genotypes so that each LD cluster contributes one test rather than one test per constituent marker. Stored consensus genotypes cover clusters of at least five markers; for the remaining clusters containing a sorted marker, consensus genotypes were constructed by the identical procedure, so that every cluster is tested at whatever resolution its size permits and none is represented only by its individual markers.

Testing a cluster requires parental as well as hybrid consensus genotypes, since orientation is defined from the parents. The parental consensus was constructed using the reference-marker choice and per-marker flip decisions derived from the hybrid genotypes, rather than recomputed on the parental individuals alone. Recomputing them on a small and strongly differentiated parental sample can select a different reference marker and flip a different subset of markers, which would place the hybrid and parental consensus genotypes on opposite orientations for any marker where the two choices disagree, without any error being raised.

The diagnostic index of a cluster was taken as the maximum across its member markers, so that a cluster is excluded by a DI criterion only if none of its members is ancestry-informative. Using the representative marker alone would exclude clusters whose informative markers lie elsewhere in the cluster.

S4.6 Distinguishing absence of signal from dilution

A cluster may contain a marker that sorts individually and yet not sort once its members are aggregated. Two situations produce this, and they were separated using the same round-trip fidelity measure that governs merging: a coherent cluster whose consensus is faithful and genuinely does not sort, versus a heterogeneous cluster in which averaging across dissimilar members dilutes a real signal. Blocks falling below the fidelity floor used during clustering were flagged for separate consideration.

S5 Genomic architecture

S5.1 Genomic covariates

Recombination rate was assigned per marker by linear interpolation of the linkage map, in centimorgans per megabase. From the parental allele frequencies p_{aqu} and p_{pol} we computed within-species diversity as the mean of the two species’ expected heterozygosities, absolute divergence as

$$d_{xy} = p_{\text{aqu}}(1 - p_{\text{pol}}) + p_{\text{pol}}(1 - p_{\text{aqu}}), \quad (5)$$

and relative differentiation F_{ST} in the standard form $(H_T - H_S)/H_T$. Reporting relative and absolute measures alongside diversity is what allows a peak of differentiation caused by reduced within-species diversity to be distinguished from one caused by genuinely elevated divergence, a distinction that a relative measure alone cannot make.

S5.2 Marker-level versus unit-level summaries

Sorting was summarised across recombination deciles twice: once treating each independent unit as one observation, using the pruning representatives, and once on a random sample of individual markers. The comparison is diagnostic rather than decorative. Adjacent markers within a low-recombination region report the same underlying event, so a relationship that is present at marker level and absent at unit level reflects the spatial redundancy of markers rather than a property of the units themselves.

S5.3 Cluster size as collider and as confounder

Block size occupies a different causal position in different questions here, and is handled differently in each; we state this explicitly because the same variable is deliberately treated in opposite ways.

When relating recombination to sorting, cluster size is a consequence of both recombination and local diversity, and therefore a collider. It is excluded from those models: conditioning on it induces an apparent association between recombination and sorting that is not otherwise present.

When relating climate-association outlier status to the diagnostic index (Section S6), cluster size is a confounder. It independently predicts DI genome-wide, and it is also required by any criterion that counts significant members within a cluster. It is therefore adjusted for in those models.

S5.4 Direction of sorting versus architecture

To ask whether genomic architecture predicts the *direction* of sorting beyond the diagnostic index, the direction of unidirectionally sorted units was modelled by logistic regression on standardised diagnostic index, log recombination rate, pooled parental allele frequency and log cluster size, fitted across units. Any recombination term here is an association: under neutrality a non-recombining autosomal region does not experience stronger drift than its surroundings, so attributing such a term to linked selection requires an explicit demographic null, which is evaluated separately.

S6 Association with climate

S6.1 Genotype–environment association

Per-population allele frequencies were associated with per-population climate covariates—two principal components of the climate variables, constant within a population—using BayPass, with

evidence reported as Bayes factors on the deciban scale. Eight runs were performed, crossing two population sets, the two climate axes, and two treatments of the among-population covariance matrix. In one treatment the covariance matrix is estimated beforehand from the LD-pruned marker set under the core model and supplied to the association run; in the other it is estimated internally from the full marker set. Estimating it from the pruned set is preferable in principle, because linkage disequilibrium among markers otherwise inflates the apparent covariance among populations; running both allows the sensitivity of any result to that choice to be reported.

The association output carries no marker identifier, only a row index into the genotype file supplied to the program. Results were therefore matched to markers positionally, and every import is guarded by assertions that the number of rows equals the number of markers and that the index is a strictly increasing sequence, so that a mismatch caused by regenerating the inputs raises an error rather than silently assigning evidence to the wrong markers.

S6.2 Ancestry as a potential confounder

Sorting is by definition the fixation of parental ancestry, so a climate axis that is itself correlated with genome-wide ancestry could appear to co-localise with sorting for structural rather than ecological reasons. We therefore estimated ancestry directly: for each population, the mean *F. aquilonia*-allele frequency across ancestry-diagnostic markers, oriented from the parental samples, and examined its correlation with each climate axis. Because a correlation across twenty populations can rest on very few of them, the correlation was recomputed with individual populations removed to identify any that carried it.

Two controls follow from this and are applied throughout. First, conditioning on the among-population covariance matrix absorbs the shared population structure with which such an axis is correlated. Second, the population set excluding the population identified as carrying the ancestry-climate correlation was taken as the primary analysis, with the full set retained as a comparator. Results are reported across all eight runs so that dependence on either choice is visible.

S6.3 Association strength per cluster

Each cluster’s climate association was summarised as its *significance rate*: the proportion of its member markers reaching the Bayes factor threshold. A rate is used rather than a count of significant members because a count makes eligibility depend on cluster size. A cluster of nine markers would need more than half of its members to be individually significant in order to reach a count of five, whereas a cluster of ninety needs only a small fraction; the same nominal criterion is therefore far stricter for small clusters. This cannot be repaired by conditioning on cluster size, because the resulting eligibility is a hard, strongly non-linear function of size rather than a smooth trend, and the adjustment would have to extrapolate across a range in which one outcome cannot occur at all. Expressing association as a rate removes the dependence by construction, and cluster size is retained as a covariate for its independent effects.

Because a rate is estimated imprecisely when a cluster contains few markers, results are reported across nested size strata rather than pooled: all clusters, those with at least twenty markers, and those with at least fifty. Reporting the strata together makes the dependence of any estimate on measurement precision in the predictor explicit.

S6.4 Testing at the level of clusters

Blocks are the units of observation, as elsewhere. For the diagnostic-index outcome, the quantity modelled is the proportion of a cluster’s members that are ancestry-informative, with each cluster

contributing one observation. A size-weighted model with an overdispersion correction is reported alongside it as a check. The corresponding marker-level model is not used as the primary analysis: it treats markers within a cluster as independent trials, whereas they are in linkage disequilibrium by construction, which understates the standard errors substantially. The relationship between climate association and the sorting classification was tested in the same way, with the cluster's classification as the outcome.

S7 Software and reproducibility

LD decay estimation, complexity reduction, consensus genotype construction and the associated diagnostics are implemented in the **LDscnR** package; the analyses described above are scripted in the project repository, one script per stage.

Two conventions support reproducibility. Clustering results and any output whose content depends on an analysis threshold are written to filenames that encode the parameters used, so that alternative settings coexist rather than overwrite one another and no analysis can silently be run against a clustering it was not built on. Second, the computationally expensive intermediate—the parental consensus genotypes—is cached, and is by construction independent of the downstream thresholds, so that re-analysis under a different criterion does not require it to be recomputed.

Table S1: Parameter settings used throughout.

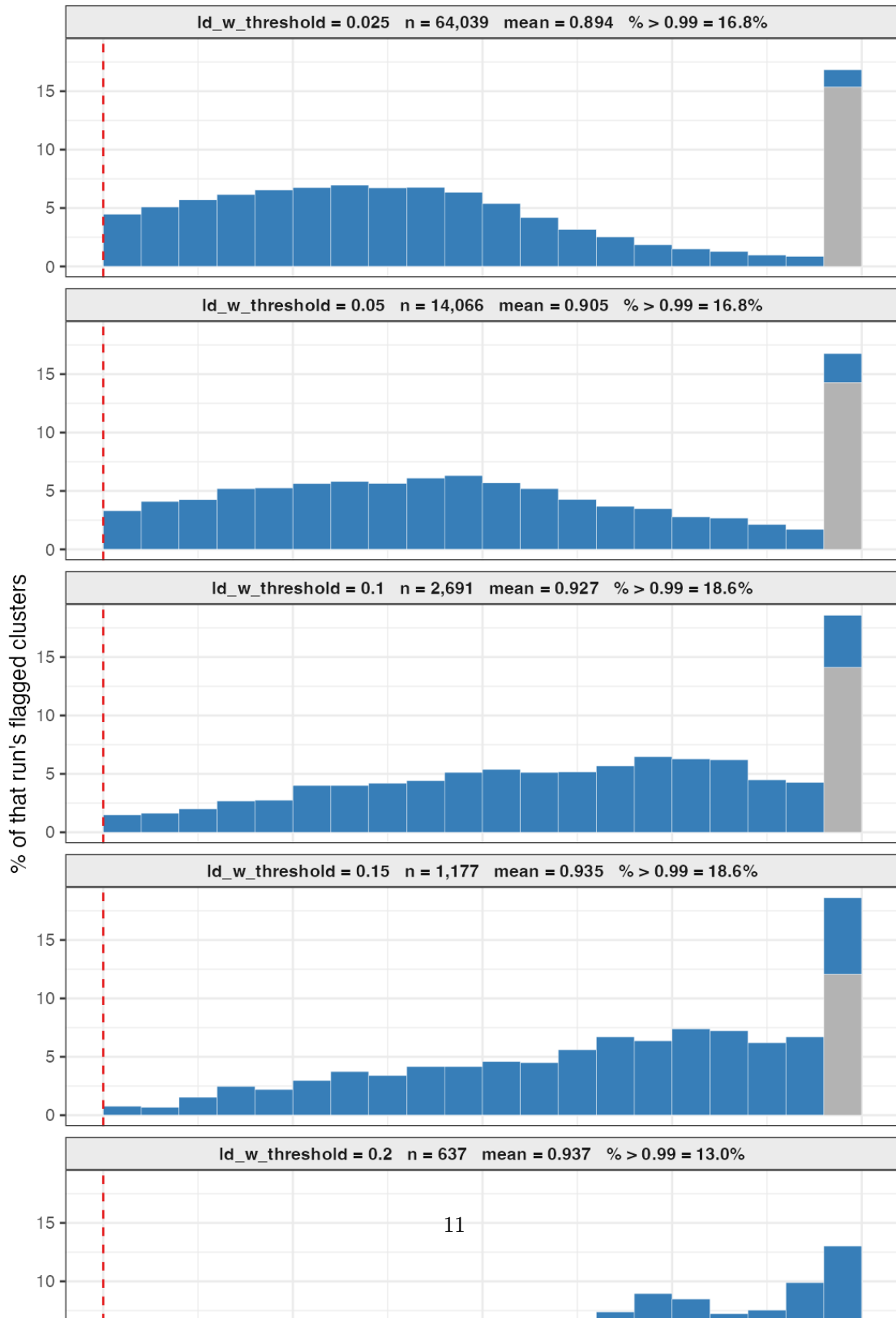
Stage	Parameter	Value
Marker set	site type	biallelic
	MAF among hybrids	≥ 0.05 (1,114,340 markers)
LD decay	background pairs	inter-chromosomal, 5,000 markers
	windows per chromosome	20, 50% overlap
	pairs per window	$\leq 5,000$, 20 distance strata
	r^2 quantile	0.95
	MAF for curve fitting	> 0.1 (both members)
	robust aggregation	central 95% of windows
	local support ld_w	median r^2 within $\rho = 0.95$ window
Stage 1	clustering threshold	per chromosome at $\rho = 0.5$
	linkage	components, then complete linkage
	representative	highest median r^2 to cluster
Stage 2	flagging	$ld_w > 0.025$ or ≥ 5 markers
	consensus fidelity floor	$score_{eMLG} \geq 0.80$
	pairwise correlation floor	$r^2 \geq 0.2$
	distance restriction	0.5 cM (genetic map)
	consensus size threshold	≥ 5 markers
Sorting	near-fixation tolerance	0.15
	classification threshold	0.5
	null probability	0.5
	minimum fixed populations	5
	parental MAF gate	≥ 0.15 (pooled counts)
	cluster diagnostic index	maximum across members
	multiple testing	Benjamini–Hochberg
Climate	evidence threshold	$BF(dB) \geq 15$
	runs	2 population sets $\times$ 2 axes $\times$ 2 covariance settings
	diagnostic-index criterion	$DI > -25$
	size strata	all, ≥ 20 , ≥ 50 markers

Stage 2 (dynamic cut) score distribution by Id_w_threshold

Dashed line = score_threshold (0.8), the floor every accepted merge must clear.

Grey = singleton clusters (flagged but never merged); blue = genuine multi-locus merges.

merged (n_loci > 1) singleton (n_loci = 1, no merge possible)



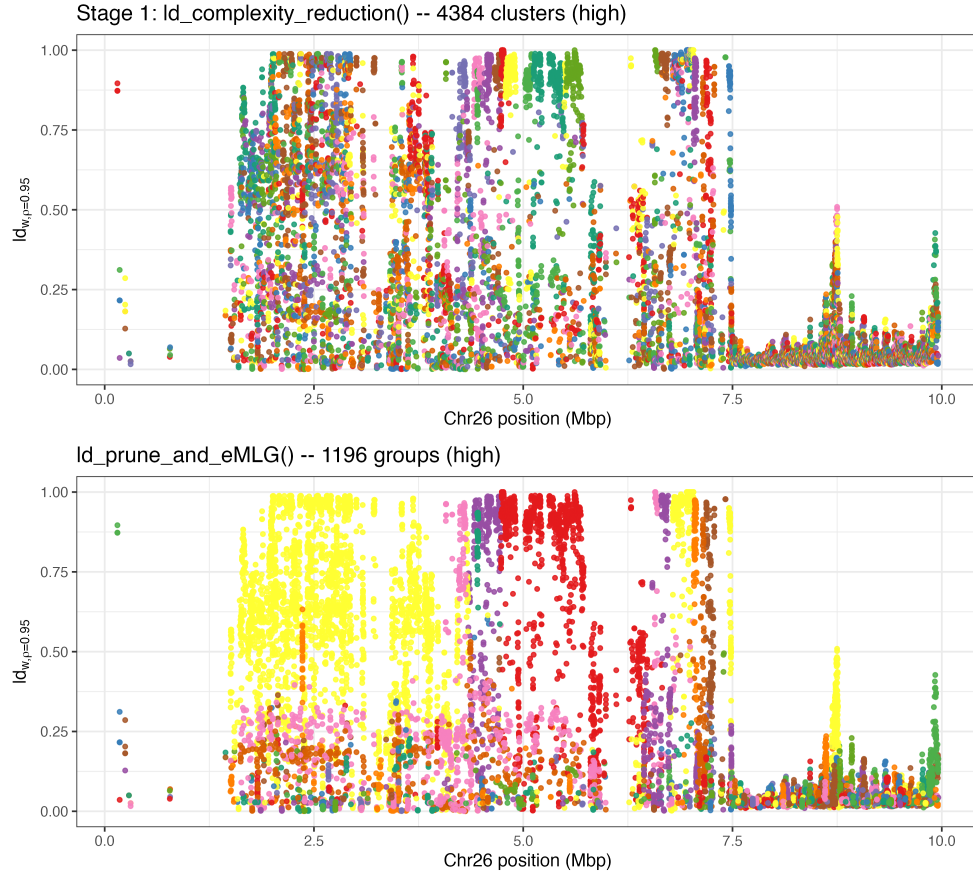


Figure S2: Stage 1 clustering compared with the combined two-stage result for Chr26, restricted to markers with high local LD support. Fragmented Stage-1 clusters within each low-recombination region are consolidated into single coherent clusters per physical LD peak, which is the intended effect of Stage 2.